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In [1]: # 1. Install the core machine learning libraries from the standard repository  
!pip install -U transformers accelerate bitsandbytes datasets  
  
# Force-reinstall only the core math/vision frameworks from the correct CUDA 12  
!pip install --force-reinstall torch torchvision torchaudio --index-url https://
```

```

WARNING: Ignoring invalid distribution -vidia-cusparse-cu12 (/opt/conda/envs/pytorch/lib/python3.10/site-packages)
Requirement already satisfied: transformers in /opt/conda/envs/pytorch/lib/python3.10/site-packages (5.14.1)
Collecting transformers
  Downloading transformers-5.15.1-py3-none-any.whl.metadata (32 kB)
Requirement already satisfied: accelerate in /opt/conda/envs/pytorch/lib/python3.10/site-packages (1.14.0)
Requirement already satisfied: bitsandbytes in /opt/conda/envs/pytorch/lib/python3.10/site-packages (0.49.2)
Collecting bitsandbytes
  Downloading bitsandbytes-0.50.1-py3-none-manylinux_2_24_x86_64.whl.metadata (10 kB)
Requirement already satisfied: datasets in /opt/conda/envs/pytorch/lib/python3.10/site-packages (5.0.0)
Collecting datasets
  Downloading datasets-5.0.1-py3-none-any.whl.metadata (23 kB)
Requirement already satisfied: huggingface-hub<2.0,>=1.5.0 in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from transformers) (1.22.0)
Requirement already satisfied: numpy>=1.17 in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from transformers) (2.2.6)
Requirement already satisfied: packaging>=20.0 in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from transformers) (26.2)
Requirement already satisfied: pyyaml>=5.1 in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from transformers) (6.0.3)
Requirement already satisfied: regex>=2025.10.22 in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from transformers) (2026.6.28)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from transformers) (0.22.2)
Requirement already satisfied: typer in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from transformers) (0.26.8)
Requirement already satisfied: safetensors>=0.8.0 in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from transformers) (0.8.0)
Requirement already satisfied: tqdm>=4.60 in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from transformers) (4.68.3)
Requirement already satisfied: click<9.0.0,>=8.4.2 in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (8.4.2)
Requirement already satisfied: filelock>=3.10.0 in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (3.29.0)
Requirement already satisfied: fsspec>=2023.5.0 in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (2026.4.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.5.1 in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (1.5.1)
Requirement already satisfied: httpx<1,>=0.23.0 in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (0.28.1)
Requirement already satisfied: typing-extensions>=4.1.0 in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (4.15.0)
Requirement already satisfied: anyio in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transformers) (4.14.0)
Requirement already satisfied: certifi in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transformers) (2026.6.17)
Requirement already satisfied: httpcore==1.* in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transformers) (1.0.9)

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Requirement already satisfied: idna in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transformers) (3.18)

Requirement already satisfied: h11<=0.16 in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transformers) (0.16.0)

Requirement already satisfied: psutil in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from accelerate) (5.9.3)

Requirement already satisfied: torch>=2.0.0 in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from accelerate) (2.13.0+cu126)

Requirement already satisfied: setuptools>=77.0.3 in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from torch>=2.0.0->accelerate) (78.1.0)

Requirement already satisfied: sympy>=1.13.3 in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from torch>=2.0.0->accelerate) (1.14.0)

Requirement already satisfied: networkx>=2.5.1 in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from torch>=2.0.0->accelerate) (3.4.2)

Requirement already satisfied: jinja2 in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from torch>=2.0.0->accelerate) (3.1.6)

Requirement already satisfied: cuda-toolkit==12.6.3 in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.6.3; platform_system == "Linux"->torch>=2.0.0->accelerate) (12.6.3)

Requirement already satisfied: cuda-bindings<13,>=12.9.4 in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from torch>=2.0.0->accelerate) (12.9.4)

Requirement already satisfied: nvidia-cudnn-cu12==9.10.2.21 in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from torch>=2.0.0->accelerate) (9.10.2.21)

Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from torch>=2.0.0->accelerate) (0.7.1)

Requirement already satisfied: nvidia-nccl-cu12==2.29.3 in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from torch>=2.0.0->accelerate) (2.29.3)

Requirement already satisfied: nvidia-nvshmem-cu12==3.4.5 in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from torch>=2.0.0->accelerate) (3.4.5)

Requirement already satisfied: triton==3.7.1 in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from torch>=2.0.0->accelerate) (3.7.1)

Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.6.85.* in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.6.3; platform_system == "Linux"->torch>=2.0.0->accelerate) (12.6.85)

Requirement already satisfied: nvidia-cusparselt-cu12==12.5.4.2.* in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.6.3; platform_system == "Linux"->torch>=2.0.0->accelerate) (12.5.4.2)

Requirement already satisfied: nvidia-cuda-runtime-cu12==12.6.77.* in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.6.3; platform_system == "Linux"->torch>=2.0.0->accelerate) (12.6.77)

Requirement already satisfied: nvidia-cuda-cupti-cu12==12.6.80.* in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.6.3; platform_system == "Linux"->torch>=2.0.0->accelerate) (12.6.80)

Requirement already satisfied: nvidia-nvtx-cu12==12.6.77.* in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.6.3; platform_system == "Linux"->torch>=2.0.0->accelerate) (12.6.77)

Requirement already satisfied: nvidia-cublas-cu12==12.6.4.1.* in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.6.3; platform_system == "Linux"->torch>=2.0.0->accelerate) (12.6.4.1)

Requirement already satisfied: nvidia-curand-cu12==10.3.7.77.* in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.6.3; platform_system == "Linux"->torch>=2.0.0->accelerate) (10.3.7.77)

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ile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.6.3; platform_system
== "Linux"->torch>=2.0.0->accelerate) (10.3.7.77)
Requirement already satisfied: nvidia-cusolver-cu12==11.7.1.2.* in /opt/conda/env
s/pytorch/lib/python3.10/site-packages (from cuda-toolkit[cublas,cudart,cufft,cuf
ile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.6.3; platform_system
== "Linux"->torch>=2.0.0->accelerate) (11.7.1.2)
Requirement already satisfied: nvidia-nvjitlink-cu12==12.6.85.* in /opt/conda/env
s/pytorch/lib/python3.10/site-packages (from cuda-toolkit[cublas,cudart,cufft,cuf
ile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.6.3; platform_system
== "Linux"->torch>=2.0.0->accelerate) (12.6.85)
Requirement already satisfied: nvidia-cufft-cu12==11.3.0.4.* in /opt/conda/envs/p
ytorch/lib/python3.10/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufil
e,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.6.3; platform_system =
= "Linux"->torch>=2.0.0->accelerate) (11.3.0.4)
Requirement already satisfied: nvidia-cufile-cu12==1.11.1.6.* in /opt/conda/envs/
pytorch/lib/python3.10/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufil
e,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.6.3; platform_system =
= "Linux"->torch>=2.0.0->accelerate) (1.11.1.6)
Requirement already satisfied: cuda-pathfinder~=1.1 in /opt/conda/envs/pytorch/li
b/python3.10/site-packages (from cuda-bindings<13,>=12.9.4->torch>=2.0.0->acceler
ate) (1.2.2)
Requirement already satisfied: pyarrow>=21.0.0 in /opt/conda/envs/pytorch/lib/pyt
hon3.10/site-packages (from datasets) (24.0.0)
Requirement already satisfied: dill<0.4.2,>=0.3.0 in /opt/conda/envs/pytorch/lib/
python3.10/site-packages (from datasets) (0.4.1)
Requirement already satisfied: pandas in /opt/conda/envs/pytorch/lib/python3.10/s
ite-packages (from datasets) (2.3.3)
Requirement already satisfied: requests>=2.32.2 in /opt/conda/envs/pytorch/lib/py
thon3.10/site-packages (from datasets) (2.34.2)
Requirement already satisfied: xxhash in /opt/conda/envs/pytorch/lib/python3.10/s
ite-packages (from datasets) (3.8.1)
Requirement already satisfied: multiprocessing<0.70.20 in /opt/conda/envs/pytorch/li
b/python3.10/site-packages (from datasets) (0.70.19)
Requirement already satisfied: aiohttp!=4.0.0a0,!4.0.0a1 in /opt/conda/envs/pyto
rch/lib/python3.10/site-packages (from fsspec[http]<=2026.6.0,>=2023.1.0->dataset
s) (3.14.1)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /opt/conda/envs/pytorc
h/lib/python3.10/site-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=20
26.6.0,>=2023.1.0->datasets) (2.6.2)
Requirement already satisfied: aiosignal>=1.4.0 in /opt/conda/envs/pytorch/lib/py
thon3.10/site-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2026.6.0,>
=2023.1.0->datasets) (1.4.0)
Requirement already satisfied: async-timeout<6.0,>=4.0 in /opt/conda/envs/pytorc
h/lib/python3.10/site-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=20
26.6.0,>=2023.1.0->datasets) (5.0.1)
Requirement already satisfied: attrs>=17.3.0 in /opt/conda/envs/pytorch/lib/pytho
n3.10/site-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2026.6.0,>=20
23.1.0->datasets) (26.1.0)
Requirement already satisfied: frozenlist>=1.1.1 in /opt/conda/envs/pytorch/lib/p
ython3.10/site-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2026.6.0,
>=2023.1.0->datasets) (1.8.0)
Requirement already satisfied: multidict<7.0,>=4.5 in /opt/conda/envs/pytorch/li
b/python3.10/site-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2026.
6.0,>=2023.1.0->datasets) (6.7.1)
Requirement already satisfied: propcache>=0.2.0 in /opt/conda/envs/pytorch/lib/py
thon3.10/site-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2026.6.0,>
=2023.1.0->datasets) (0.5.2)
Requirement already satisfied: yarll<2.0,>=1.17.0 in /opt/conda/envs/pytorch/lib/p
ython3.10/site-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2026.6.0,
>=2023.1.0->datasets) (1.24.2)

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Requirement already satisfied: charset_normalizer<4,>=2 in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from requests>=2.32.2->datasets) (3.4.7)
Requirement already satisfied: urllib3<3,>=1.26 in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from requests>=2.32.2->datasets) (2.7.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from sympy>=1.13.3->torch>=2.0.0->accelerate) (1.3.0)
Requirement already satisfied: exceptiongroup>=1.0.2 in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from anyio->httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transformers) (1.3.1)
Requirement already satisfied: MarkupSafe>=2.0 in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from jinja2->torch>=2.0.0->accelerate) (3.0.3)
Requirement already satisfied: python-dateutil>=2.8.2 in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from pandas->datasets) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from pandas->datasets) (2026.2)
Requirement already satisfied: tzdata>=2022.7 in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from pandas->datasets) (2026.2)
Requirement already satisfied: six>=1.5 in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from python-dateutil>=2.8.2->pandas->datasets) (1.17.0)
Requirement already satisfied: shellingham>=1.3.0 in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from typer->transformers) (1.5.4)
Requirement already satisfied: rich>=13.8.0 in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from typer->transformers) (13.9.4)
Requirement already satisfied: annotated-doc>=0.0.2 in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from typer->transformers) (0.0.4)
Requirement already satisfied: markdown-it-py>=2.2.0 in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from rich>=13.8.0->typer->transformers) (4.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from rich>=13.8.0->typer->transformers) (2.20.0)
Requirement already satisfied: mdurl~=0.1 in /opt/conda/envs/pytorch/lib/python3.10/site-packages (from markdown-it-py>=2.2.0->rich>=13.8.0->typer->transformers) (0.1.2)
Downloading transformers-5.15.1-py3-none-any.whl (11.7 MB)
_____ 11.7/11.7 MB 138.9 MB/s 0:00:00
Downloading bitsandbytes-0.50.1-py3-none-manylinux_2_24_x86_64.whl (41.0 MB)
_____ 41.0/41.0 MB 117.6 MB/s 0:00:00 eta
0:00:01
Downloading datasets-5.0.1-py3-none-any.whl (559 kB)
_____ 559.1/559.1 kB 40.2 MB/s 0:00:00
WARNING: Ignoring invalid distribution -vidia-cusparse-cu12 (/opt/conda/envs/pytorch/lib/python3.10/site-packages)
Installing collected packages: datasets, bitsandbytes, transformers
  Attempting uninstall: datasets
    Found existing installation: datasets 5.0.0
    Uninstalling datasets-5.0.0:
      Successfully uninstalled datasets-5.0.0
  Attempting uninstall: bitsandbytes
    Found existing installation: bitsandbytes 0.49.2
    Uninstalling bitsandbytes-0.49.2:
      Successfully uninstalled bitsandbytes-0.49.2
  Attempting uninstall: transformers
    Found existing installation: transformers 5.14.1
    Uninstalling transformers-5.14.1:
      Successfully uninstalled transformers-5.14.1
  Successfully installed bitsandbytes-0.50.1 datasets-5.0.1 transformers-5.15.1

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WARNING: Ignoring invalid distribution -vidia-cusparselt-cu12 (/opt/conda/envs/pytorch/lib/python3.10/site-packages)
Looking in indexes: https://download.pytorch.org/whl/cu126
Collecting torch
  Using cached torch-2.13.0%2Bcu126-cp310-cp310-manylinux_2_28_x86_64.whl.metadata (38 kB)
Collecting torchvision
  Using cached torchvision-0.28.0%2Bcu126-cp310-cp310-manylinux_2_28_x86_64.whl.metadata (5.6 kB)
Collecting torchaudio
  Using cached torchaudio-2.11.0%2Bcu126-cp310-cp310-manylinux_2_28_x86_64.whl.metadata (6.9 kB)
Collecting filelock (from torch)
  Downloading filelock-3.32.3-py3-none-any.whl.metadata (2.0 kB)
Collecting typing-extensions>=4.10.0 (from torch)
  Downloading typing_extensions-4.16.0-py3-none-any.whl.metadata (3.3 kB)
Collecting setuptools>=77.0.3 (from torch)
  Using cached setuptools-78.1.0-py3-none-any.whl.metadata (6.6 kB)
Collecting sympy>=1.13.3 (from torch)
  Using cached sympy-1.14.0-py3-none-any.whl.metadata (12 kB)
Collecting networkx>=2.5.1 (from torch)
  Using cached networkx-3.4.2-py3-none-any.whl.metadata (6.3 kB)
Collecting jinja2 (from torch)
  Using cached jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)
Collecting fsspec>=0.8.5 (from torch)
  Downloading fsspec-2026.7.0-py3-none-any.whl.metadata (10 kB)
Collecting cuda-toolkit==12.6.3 (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsen,nvjitlink,nvrtc,nvtx]==12.6.3; platform_system == "Linux"->torch)
  Downloading cuda_toolkit-12.6.3-py2.py3-none-any.whl (2.3 kB)
Collecting cuda-bindings<13,>=12.9.4 (from torch)
  Downloading cuda_bindings-12.9.7-cp310-cp310-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata (2.6 kB)
Collecting nvidia-cudnn-cu12==9.10.2.21 (from torch)
  Downloading nvidia_cudnn_cu12-9.10.2.21-py3-none-manylinux_2_27_x86_64.whl (706.8 MB)
  ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 706.8/706.8 MB 323.3 MB/s 0:00:020
0:0100:01
Collecting nvidia-cusparselt-cu12==0.7.1 (from torch)
  Downloading nvidia_cusparselt_cu12-0.7.1-py3-none-manylinux2014_x86_64.whl (287.2 MB)
  ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 287.2/287.2 MB 238.9 MB/s 0:00:01a
0:00:01
Collecting nvidia-nccl-cu12==2.29.3 (from torch)
  Downloading nvidia_nccl_cu12-2.29.3-py3-none-manylinux_2_18_x86_64.whl (289.8 MB)
  ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 289.8/289.8 MB 302.8 MB/s 0:00:010
0:0100:01
Collecting nvidia-nvshmem-cu12==3.4.5 (from torch)
  Downloading nvidia_nvshmem_cu12-3.4.5-py3-none-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (139.1 MB)
  ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 139.1/139.1 MB 356.9 MB/s 0:00:00a
0:00:01
Collecting triton==3.7.1 (from torch)
  Using cached triton-3.7.1-cp310-cp310-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (1.7 kB)
Collecting nvidia-cusolver-cu12==11.7.1.2.* (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsen,nvjitlink,nvrtc,nvtx]==12.6.3; platform_system == "Linux"->torch)
  Downloading nvidia_cusolver_cu12-11.7.1.2-py3-none-manylinux2014_x86_64.manylin

```

ux_2_17_x86_64.whl (158.2 MB)

158.2/158.2 MB 291.2 MB/s 0:00:00a
0:00:01

Collecting nvidia-nvtx-cu12==12.6.77.* (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.6.3; platform_system == "Linux"->torch)

Downloading nvidia_nvtx_cu12-12.6.77-py3-none-manylinux2014_x86_64.whl (89 kB)

Collecting nvidia-cusparsen-cu12==12.5.4.2.* (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.6.3; platform_system == "Linux"->torch)

Downloading nvidia_cusparsen_cu12-12.5.4.2-py3-none-manylinux2014_x86_64.whl (216.6 MB)

216.6/216.6 MB 297.1 MB/s 0:00:00a
0:00:01

Collecting nvidia-cuda-nvrtc-cu12==12.6.85.* (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.6.3; platform_system == "Linux"->torch)

Downloading nvidia_cuda_nvrtc_cu12-12.6.85-py3-none-manylinux2010_x86_64.whl (23.6 MB)

23.6/23.6 MB 257.7 MB/s 0:00:00

Collecting nvidia-cublas-cu12==12.6.4.1.* (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.6.3; platform_system == "Linux"->torch)

Downloading nvidia_cublas_cu12-12.6.4.1-py3-none-manylinux2014_x86_64.whl (393.1 MB)

393.1/393.1 MB 322.8 MB/s 0:00:010
0:0100:01

Collecting nvidia-nvjitlink-cu12==12.6.85.* (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.6.3; platform_system == "Linux"->torch)

Downloading nvidia_nvjitlink_cu12-12.6.85-py3-none-manylinux2010_x86_64.whl (19.7 MB)

19.7/19.7 MB 336.2 MB/s 0:00:00

Collecting nvidia-cuda-cupti-cu12==12.6.80.* (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.6.3; platform_system == "Linux"->torch)

Downloading nvidia_cuda_cupti_cu12-12.6.80-py3-none-manylinux2014_x86_64.whl (8.9 MB)

8.9/8.9 MB 337.3 MB/s 0:00:00

Collecting nvidia-cufft-cu12==11.3.0.4.* (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.6.3; platform_system == "Linux"->torch)

Downloading nvidia_cufft_cu12-11.3.0.4-py3-none-manylinux2014_x86_64.whl (200.2 MB)

200.2/200.2 MB 300.1 MB/s 0:00:000
0:0100:01

Collecting nvidia-cufile-cu12==1.11.1.6.* (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.6.3; platform_system == "Linux"->torch)

Downloading nvidia_cufile_cu12-1.11.1.6-py3-none-manylinux2014_x86_64.whl (1.1 MB)

1.1/1.1 MB 333.5 MB/s 0:00:00

Collecting nvidia-cuda-runtime-cu12==12.6.77.* (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.6.3; platform_system == "Linux"->torch)

Downloading nvidia_cuda_runtime_cu12-12.6.77-py3-none-manylinux2014_x86_64.whl (897 kB)

897.7/897.7 kB 317.9 MB/s 0:00:00

Collecting nvidia-curand-cu12==10.3.7.77.* (from cuda-toolkit[cublas,cudart,cufft

```

t,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.6.3; platform_
system == "Linux"->torch)
  Downloading nvidia_curand_cu12-10.3.7.77-py3-none-manylinux2014_x86_64.manylinu
x_2_17_x86_64.whl (56.3 MB)
  _____ 56.3/56.3 MB 363.8 MB/s 0:00:00
Collecting cuda-pathfinder~=1.1 (from cuda-bindings<13,>=12.9.4->torch)
  Downloading cuda_pathfinder-1.6.0-py3-none-any.whl.metadata (1.9 kB)
Collecting numpy (from torchvision)
  Using cached numpy-2.2.6-cp310-cp310-manylinux_2_17_x86_64.manylinux2014_x86_6
4.whl.metadata (62 kB)
Collecting pillow!=8.3.*,>=5.3.0 (from torchvision)
  Downloading pillow-12.3.0-cp310-cp310-manylinux_2_27_x86_64.manylinux_2_28_x86_
64.whl.metadata (9.1 kB)
Collecting mpmath<1.4,>=1.1.0 (from sympy>=1.13.3->torch)
  Using cached mpmath-1.3.0-py3-none-any.whl.metadata (8.6 kB)
Collecting MarkupSafe>=2.0 (from jinja2->torch)
  Using cached markupsafe-3.0.3-cp310-cp310-manylinux2014_x86_64.manylinux_2_17_x
86_64.manylinux_2_28_x86_64.whl.metadata (2.7 kB)
Using cached torch-2.13.0%2Bcu126-cp310-cp310-manylinux_2_28_x86_64.whl (843.7 M
B)
Using cached triton-3.7.1-cp310-cp310-manylinux_2_27_x86_64.manylinux_2_28_x86_6
4.whl (197.0 MB)
Downloading cuda_bindings-12.9.7-cp310-cp310-manylinux_2_24_x86_64.manylinux_2_28
_x86_64.whl (7.7 MB)
  _____ 7.7/7.7 MB 8.6 MB/s 0:00:0036m0:00:0
1
Downloading cuda_pathfinder-1.6.0-py3-none-any.whl (54 kB)
Using cached torchvision-0.28.0%2Bcu126-cp310-cp310-manylinux_2_28_x86_64.whl (7.
5 MB)
Using cached torchaudio-2.11.0%2Bcu126-cp310-cp310-manylinux_2_28_x86_64.whl (1.5
MB)
Downloading fsspec-2026.7.0-py3-none-any.whl (206 kB)
Using cached networkx-3.4.2-py3-none-any.whl (1.7 MB)
Downloading pillow-12.3.0-cp310-cp310-manylinux_2_27_x86_64.manylinux_2_28_x86_6
4.whl (6.9 MB)
  _____ 6.9/6.9 MB 6.9 MB/s 0:00:0036m0:00:0
1
Using cached setuptools-78.1.0-py3-none-any.whl (1.3 MB)
Using cached sympy-1.14.0-py3-none-any.whl (6.3 MB)
Using cached mpmath-1.3.0-py3-none-any.whl (536 kB)
Downloading typing_extensions-4.16.0-py3-none-any.whl (45 kB)
Downloading filelock-3.32.3-py3-none-any.whl (98 kB)
Using cached jinja2-3.1.6-py3-none-any.whl (134 kB)
Using cached markupsafe-3.0.3-cp310-cp310-manylinux2014_x86_64.manylinux_2_17_x86
_64.manylinux_2_28_x86_64.whl (20 kB)
Using cached numpy-2.2.6-cp310-cp310-manylinux_2_17_x86_64.manylinux2014_x86_64.w
hl (16.8 MB)
WARNING: Ignoring invalid distribution -vidia-cusparselt-cu12 (/opt/conda/envs/py
torch/lib/python3.10/site-packages)
Installing collected packages: torchaudio, nvidia-cusparselt-cu12, mpmath, cuda-t
oolkit, typing-extensions, triton, sympy, setuptools, pillow, nvidia-nvtx-cu12, n
vidia-nvshmem-cu12, nvidia-nvjitlink-cu12, nvidia-nccl-cu12, nvidia-curand-cu12,
nvidia-cufile-cu12, nvidia-cuda-runtime-cu12, nvidia-cuda-nvrtc-cu12, nvidia-cuda
-cupti-cu12, nvidia-cublas-cu12, numpy, networkx, MarkupSafe, fsspec, filelock, c
uda-pathfinder, nvidia-cusparselt-cu12, nvidia-cufft-cu12, nvidia-cudnn-cu12, jinja
2, cuda-bindings, nvidia-cusolver-cu12, torch, torchvision
Attempting uninstall: torchaudio
  Found existing installation: torchaudio 2.11.0+cu126
  Uninstalling torchaudio-2.11.0+cu126:
    Successfully uninstalled torchaudio-2.11.0+cu126

```

```

Attempting uninstall: nvidia-cusparse-cu12 0/33 [torchaudio]
  Found existing installation: nvidia-cusparse-cu12 0.7.10m [torchaudio]
  Uninstalling nvidia-cusparse-cu12-0.7.1:0m 0/33 [torchaudio]
  Successfully uninstalled nvidia-cusparse-cu12-0.7.1 [torchaudio]
Attempting uninstall: mpmath 1/33 [nvidia-cusparse-cu12]
  Found existing installation: mpmath 1.3.0 1/33 [nvidia-cusparse-cu12]
  Uninstalling mpmath-1.3.0: 1/33 [nvidia-cusparse-cu12]
  Successfully uninstalled mpmath-1.3.0 1/33 [nvidia-cusparse-cu12]
Attempting uninstall: cuda-toolkit 2/33 [mpmath]
  Found existing installation: cuda-toolkit 12.6.3 2/33 [mpmath]
  Uninstalling cuda-toolkit-12.6.3: 2/33 [mpmath]
  Successfully uninstalled cuda-toolkit-12.6.3 2/33 [mpmath]
Attempting uninstall: typing-extensions 2/33 [mpmath]
  Found existing installation: typing_extensions 4.15.0 2/33 [mpmath]
  Uninstalling typing_extensions-4.15.0: 2/33 [mpmath]
  Successfully uninstalled typing_extensions-4.15.0 4/33 [typing-extensions]
Attempting uninstall: triton 4/33 [typing-extensions]
  Found existing installation: triton 3.7.1 4/33 [typing-extensions]
  Uninstalling triton-3.7.1: 4/33 [typing-extensions]
  Successfully uninstalled triton-3.7.1 4/33 [typing-extensions]
Attempting uninstall: sympy 5/33 [triton]
  Found existing installation: sympy 1.14.0 5/33 [triton]
  Uninstalling sympy-1.14.0: 6/33 [sympy]
  Successfully uninstalled sympy-1.14.0 6/33 [sympy]
Attempting uninstall: setuptools 6/33 [sympy]
  Found existing installation: setuptools 78.1.0 6/33 [sympy]
  Uninstalling setuptools-78.1.0: 6/33 [sympy]
  Successfully uninstalled setuptools-78.1.0 7/33 [setuptools]
Attempting uninstall: pillow 7/33 [setuptools]
  Found existing installation: pillow 12.2.0 7/33 [setuptools]
  Uninstalling pillow-12.2.0: 8/33 [pillow]
  Successfully uninstalled pillow-12.2.0 8/33 [pillow]
Attempting uninstall: nvidia-nvtx-cu12 8/33 [pillow]
  Found existing installation: nvidia-nvtx-cu12 12.6.77 8/33 [pillow]
  Uninstalling nvidia-nvtx-cu12-12.6.77: 8/33 [pillow]
  Successfully uninstalled nvidia-nvtx-cu12-12.6.77 8/33 [pillow]
Attempting uninstall: nvidia-nvshmem-cu12 8/33 [pillow]
  Found existing installation: nvidia-nvshmem-cu12 3.4.5 8/33 [pillow]
  Uninstalling nvidia-nvshmem-cu12-3.4.5: 8/33 [pillow]
  Successfully uninstalled nvidia-nvshmem-cu12-3.4.5 8/33 [pillow]
Attempting uninstall: nvidia-nvjitlink-cu12 10/33 [nvidia-nvshmem-cu12]
  Found existing installation: nvidia-nvjitlink-cu12 12.6.85 10/33 [nvidia-nvshmem-cu12]
  Uninstalling nvidia-nvjitlink-cu12-12.6.85: 10/33 [nvidia-nvshmem-cu12]
  Successfully uninstalled nvidia-nvjitlink-cu12-12.6.85 10/33 [nvidia-nvshmem-cu12]

```

```

Attempting uninstall: nvidia-nccl-cu12----- 11/33 [nvidia-
nvjitlink-cu12]
Found existing installation: nvidia-nccl-cu12 2.29.3----- 11/33 [nvidia-nvji
tlink-cu12]
Uninstalling nvidia-nccl-cu12-2.29.3:----- 11/33 [nvidia-nvji
tlink-cu12]
Successfully uninstalled nvidia-nccl-cu12-2.29.3----- 11/33 [nvidia-nvji
tlink-cu12]
Attempting uninstall: nvidia-curand-cu12----- 12/33 [nvidia-
nccl-cu12]]
Found existing installation: nvidia-curand-cu12 10.3.7.77- 12/33 [nvidia-nccl
-cu12]
Uninstalling nvidia-curand-cu12-10.3.7.77:----- 12/33 [nvidia-nccl
-cu12]
Successfully uninstalled nvidia-curand-cu12-10.3.7.77----- 12/33 [nvidia-nccl
-cu12]
Attempting uninstall: nvidia-cufile-cu12----- 13/33 [nvidia-
curand-cu12]
Found existing installation: nvidia-cufile-cu12 1.11.1.6- 13/33 [nvidia-cura
nd-cu12]
Uninstalling nvidia-cufile-cu12-1.11.1.6:----- 13/33 [nvidia-cura
nd-cu12]
Successfully uninstalled nvidia-cufile-cu12-1.11.1.6----- 13/33 [nvidia-cura
nd-cu12]
Attempting uninstall: nvidia-cuda-runtime-cu12----- 13/33 [nvidia-cura
nd-cu12]
Found existing installation: nvidia-cuda-runtime-cu12 12.6.77m 13/33 [nvidia-
curand-cu12]
Uninstalling nvidia-cuda-runtime-cu12-12.6.77:----- 13/33 [nvidia-cura
nd-cu12]
Successfully uninstalled nvidia-cuda-runtime-cu12-12.6.77[0m 13/33 [nvidia-
curand-cu12]
Attempting uninstall: nvidia-cuda-nvrtc-cu12----- 13/33 [nvidia-cura
nd-cu12]
Found existing installation: nvidia-cuda-nvrtc-cu12 12.6.85[0m 13/33 [nvidia-
curand-cu12]
Uninstalling nvidia-cuda-nvrtc-cu12-12.6.85:----- 13/33 [nvidia-cura
nd-cu12]
Successfully uninstalled nvidia-cuda-nvrtc-cu12-12.6.85- 13/33 [nvidia-cura
nd-cu12]
Attempting uninstall: nvidia-cuda-cupti-cu12----- 16/33 [nvidia-
cuda-nvrtc-cu12]
Found existing installation: nvidia-cuda-cupti-cu12 12.6.80[0m 16/33 [nvidia-
cuda-nvrtc-cu12]
Uninstalling nvidia-cuda-cupti-cu12-12.6.80:----- 16/33 [nvidia-cuda
-nvrtc-cu12]
Successfully uninstalled nvidia-cuda-cupti-cu12-12.6.80----- 17/33 [nvidia-
cuda-cupti-cu12]
Attempting uninstall: nvidia-cublas-cu12----- 17/33 [nvidia-cuda-
cupti-cu12]
Found existing installation: nvidia-cublas-cu12 12.6.4.1- 17/33 [nvidia-cuda
-cupti-cu12]
Uninstalling nvidia-cublas-cu12-12.6.4.1:----- 17/33 [nvidia-cuda
-cupti-cu12]
Successfully uninstalled nvidia-cublas-cu12-12.6.4.1----- 17/33 [nvidia-cuda
-cupti-cu12]
Attempting uninstall: numpy----- 18/33 [nvidia-cublas-cu12]
Found existing installation: numpy 2.2.6----- 18/33 [nvidia-cubl
as-cu12]
Uninstalling numpy-2.2.6:------ 18/33 [nvidia-cublas-cu12]

```

```

    Successfully uninstalled numpy-2.2.60m————— 18/33 [nvidia-cubl
as-cu12]
    Attempting uninstall: networkx[0m————— 19/33 [numpy]las-cu12]
    Found existing installation: networkx 3.4.2————— 19/33 [numpy]
    Uninstalling networkx-3.4.2:————— 20/33 [networkx]
    Successfully uninstalled networkx-3.4.20m————— 20/33 [networkx]
    Attempting uninstall: MarkupSafe0m————— 20/33 [networkx]
    Found existing installation: MarkupSafe 3.0.3————— 20/33 [networkx]
    Uninstalling MarkupSafe-3.0.3:90m————— 20/33 [networkx]
    Successfully uninstalled MarkupSafe-3.0.3————— 20/33 [networkx]
    Attempting uninstall: fsspec0m————— 20/33 [networkx]
    Found existing installation: fsspec 2026.4.0————— 20/33 [networkx]
    Uninstalling fsspec-2026.4.0:[90m————— 20/33 [networkx]
    Successfully uninstalled fsspec-2026.4.0m————— 20/33 [networkx]
    Attempting uninstall: filelock————— 22/33 [fsspec]
    Found existing installation: filelock 3.29.0m————— 22/33 [fsspec]
    Uninstalling filelock-3.29.0:m————— 22/33 [fsspec]
    Successfully uninstalled filelock-3.29.090m————— 22/33 [fsspec]
    Attempting uninstall: cuda-pathfinder————— 23/33 [filelock]
    Found existing installation: cuda-pathfinder 1.2.2————— 23/33 [filelock]
    Uninstalling cuda-pathfinder-1.2.2:m————— 23/33 [filelock]
    Successfully uninstalled cuda-pathfinder-1.2.2————— 23/33 [filelock]
    Attempting uninstall: nvidia-cuspars-cu12m————— 23/33 [filelock]
    Found existing installation: nvidia-cuspars-cu12 12.5.4.2 23/33 [filelock]
    Uninstalling nvidia-cuspars-cu12-12.5.4.2:90m————— 23/33 [filelock]
    Successfully uninstalled nvidia-cuspars-cu12-12.5.4.2— 23/33 [filelock]
    Attempting uninstall: nvidia-cufft-cu12m————— 25/33 [nvidia-cuspars-cu12]
    Found existing installation: nvidia-cufft-cu12 11.3.0.4— 25/33 [nvidia-cusp
arse-cu12]
    Uninstalling nvidia-cufft-cu12-11.3.0.4:————— 25/33 [nvidia-cuspars-cu1
2]
    Successfully uninstalled nvidia-cufft-cu12-11.3.0.4————— 25/33 [nvidia-cusp
arse-cu12]
    Attempting uninstall: nvidia-cudnn-cu120m————— 26/33 [nvidia-cufft-cu12]
    Found existing installation: nvidia-cudnn-cu12 9.10.2.21— 26/33 [nvidia-cuff
t-cu12]
    Uninstalling nvidia-cudnn-cu12-9.10.2.21:————— 26/33 [nvidia-cufft-cu12]
    Successfully uninstalled nvidia-cudnn-cu12-9.10.2.21— 26/33 [nvidia-cuff
t-cu12]
    Attempting uninstall: Jinja2————— 27/33 [nvidia-cudnn-cu12]
    Found existing installation: Jinja2 3.1.6— 27/33 [nvidia-cudnn-cu12]
    Uninstalling Jinja2-3.1.6:————— 27/33 [nvidia-cudnn-cu12]
    Successfully uninstalled Jinja2-3.1.61m————— 27/33 [nvidia-cudnn-cu12]
    Attempting uninstall: cuda-bindings[0m————— 27/33 [nvidia-cudnn-cu12]
    Found existing installation: cuda-bindings 12.9.40m————— 27/33 [nvidia-cudn
n-cu12]
    Uninstalling cuda-bindings-12.9.4:0m————— 27/33 [nvidia-cudnn-cu12]
    Successfully uninstalled cuda-bindings-12.9.4m————— 29/33 [cuda-bindings]
    Attempting uninstall: nvidia-cusolver-cu12[0m————— 29/33 [cuda-bindings]
    Found existing installation: nvidia-cusolver-cu12 11.7.1.2 29/33 [cuda-bindin
gs]
    Uninstalling nvidia-cusolver-cu12-11.7.1.2:m————— 29/33 [cuda-bindings]
    Successfully uninstalled nvidia-cusolver-cu12-11.7.1.2— 29/33 [cuda-bindin
gs]
    Attempting uninstall: torch————— 30/33 [nvidia-cusolver-cu12]
    Found existing installation: torch 2.13.0+cu126[0m————— 30/33 [nvidia-cusolver-
cu12]
    Uninstalling torch-2.13.0+cu126:————— 31/33 [torch]olver-cu12]
    Successfully uninstalled torch-2.13.0+cu126— 31/33 [torch]
    Attempting uninstall: torchvision————— 31/33 [torch]

```

```
Found existing installation: torchvision 0.28.0+cu126 31/33 [torch]
Uninstalling torchvision-0.28.0+cu126: 31/33 [torch]
Successfully uninstalled torchvision-0.28.0+cu126 31/33 [torch]
33/33 [torchvision] [torchvision]
```

ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.

datasets 5.0.1 requires fsspec[http]<=2026.6.0,>=2023.1.0, but you have fsspec 2026.7.0 which is incompatible.

WARNING: Ignoring invalid distribution -vidia-cusparse-cu12 (/opt/conda/envs/pytorch/lib/python3.10/site-packages)

Successfully installed MarkupSafe-3.0.3 cuda-bindings-12.9.7 cuda-pathfinder-1.6.0 cuda-toolkit-12.6.3 filelock-3.32.3 fsspec-2026.7.0 jinja2-3.1.6 mpmath-1.3.0 networkx-3.4.2 numpy-2.2.6 nvidia-cublas-cu12-12.6.4.1 nvidia-cuda-cupti-cu12-12.6.80 nvidia-cuda-nvrtc-cu12-12.6.85 nvidia-cuda-runtime-cu12-12.6.77 nvidia-cudnn-cu12-9.10.2.21 nvidia-cufft-cu12-11.3.0.4 nvidia-cufile-cu12-1.11.1.6 nvidia-curand-cu12-10.3.7.77 nvidia-cusolver-cu12-11.7.1.2 nvidia-cusparselt-cu12-12.5.4.2 nvidia-cusparselt-cu12-0.7.1 nvidia-nccl-cu12-2.29.3 nvidia-nvjitlink-cu12-12.6.85 nvidia-nvshmem-cu12-3.4.5 nvidia-nvtx-cu12-12.6.77 pillow-12.3.0 setuptools-78.1.0 sympy-1.14.0 torch-2.13.0+cu126 torchaudio-2.11.0+cu126 torchvision-0.28.0+cu126 triton-3.7.1 typing-extensions-4.16.0

```
In [2]: pip freeze > requirements.txt
```

WARNING: Ignoring invalid distribution -vidia-cusparse-cu12 (/opt/conda/envs/pytorch/lib/python3.10/site-packages)

Note: you may need to restart the kernel to use updated packages.

```
In [ ]: import torch
import torchvision

print("Is CUDA available:", torch.cuda.is_available())
print("Torch version:", torch.__version__)
print("Torchvision version:", torchvision.__version__)
```

```
In [ ]: from transformers import AutoProcessor, AutoModelForCausalLM

MODEL_ID = "google/gemma-4-E4B-it"

# Load model
processor = AutoProcessor.from_pretrained(MODEL_ID)
model = AutoModelForCausalLM.from_pretrained(
    MODEL_ID,
    dtype="auto",
    device_map="auto"
)
```

```
In [ ]: ## Prompt
# messages = [
#     {"role": "system", "content": "You are a helpful assistant."},
#     {"role": "user", "content": "Write a short joke about saving RAM."},
# ]

## Process input
# text = processor.apply_chat_template(
#     messages,
#     tokenize=False,
#     add_generation_prompt=True,
#     enable_thinking=False
# )
```

```
# inputs = processor(text=text, return_tensors="pt").to(model.device)
# input_len = inputs["input_ids"].shape[-1]

# # Generate output
# outputs = model.generate(**inputs, max_new_tokens=1024)
# response = processor.decode(outputs[0][input_len:], skip_special_tokens=False)

# # Parse output
# processor.parse_response(response)
```

```
In [ ]: from datasets import load_dataset

dataset = load_dataset("MathArena/aime_2026")
print(dataset)
print(len(dataset))

# # View the first problem

first_problem = dataset['train'][0]

print("Problem Statement:")
print(first_problem['problem']) # Access the 'problem' column directly

print("\nGold Answer:")
print(first_problem['answer']) # Access the 'answer' column directly
```

```
In [ ]: # messages = [
#       {"role": "system", "content": "You are an competitive math solver."},
#       {"role": "user", "content": first_problem['problem']},
#     ]

# # Apply chat template with thinking enabled
# text = processor.apply_chat_template(
#     messages,
#     tokenize=False,
#     add_generation_prompt=True,
#     enable_thinking=True
# )

# inputs = processor(text=text, return_tensors="pt").to(model.device)
# input_len = inputs["input_ids"].shape[-1]

# # Generate response
# outputs = model.generate(
#     **inputs,
#     max_new_tokens=4096,
#     temperature=1.0,
#     top_p=0.95,
#     top_k=64,
# )

# response = processor.decode(outputs[0][input_len:], skip_special_tokens=False)

# # Parse output
# processor.parse_response(response)
```

```
In [ ]: train_dataset_test = dataset["train"]
n_test = len(train_dataset_test)
```

```

for i in range(n_test):
    print(f"Problem idx: {i}")
    print(f"Problem Statement: {train_dataset_test[i]['problem']}")
    print("-" * 30)
    print(f"Problem answer: {train_dataset_test[i]['answer']}")
    print()
    print()
    print()

```

In []: *# import pandas*

```

# test_frame = pd.DataFrame(columns=[
#     "model",
#     "eval_run_idx",

```

In []: *# End of Tutorials and basic settings & sanity checks.
Previous parts contain: Setting .env in Google Agent Platform Workbench notebook
Downloading and loading the dataset from HuggingFace with the given name, with
and checking the contents of datasets.*

```

In [ ]: import os
import pandas
from pathlib import Path
from datasets import load_dataset

dataset = load_dataset("MathArena/aime_2026")
train_dataset = dataset["train"]
n = len(train_dataset)
ROOT = Path.cwd()

os.makedirs("results", exist_ok=True)
folder = ROOT / "results"

postfix = f"run1"
run_folder = folder / postfix
os.makedirs(run_folder, exist_ok=True)

for i in range(1):
    prob = train_dataset[i]["problem"]
    correct_answer = train_dataset[i]["answer"]

    messages = [
        {"role": "system", "content": "You are a math problem solver in a compet"},
        {"role": "user", "content": prob},
    ]

    # Apply chat template with thinking enabled
    text = processor.apply_chat_template(
        messages,
        tokenize=False,
        add_generation_prompt=True,
        enable_thinking=True
    )

    inputs = processor(text=text, return_tensors="pt").to(model.device)
    input_len = inputs["input_ids"].shape[-1]

```

```
# Generate response
outputs = model.generate(
    **inputs,
    max_new_tokens=10000,
    temperature=1.0,
    top_p=0.95,
    top_k=64,
)

response = processor.decode(outputs[0][input_len:], skip_special_tokens=False)

# Parse output
parsed = processor.parse_response(response)

filename = f"problem_{i}.txt"
with open(run_folder / filename, "w", encoding="utf-8") as f:
    f.write(f"Problem #{i}\n")
    f.write("=" * 80 + "\n\n")

    f.write("Problem:\n")
    f.write(prob + "\n\n")

    f.write("Correct Answer:\n")
    f.write(str(correct_answer) + "\n\n")

    f.write("Raw Response:\n")
    f.write(response + "\n\n")

    f.write("Parsed Response:\n")
    f.write(str(parsed) + "\n")
```

In []:

In []: